
Spam Email Classifier Project

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Class

Program: Artificial Intelligence

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Section: B

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Email Spam Detection - Output Report

Generated from: email_spam_detection.py

Dataset: SMS Spam Collection (spam.csv)

Terminal Output

```
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EMAIL SPAM DETECTION WITH MACHINE LEARNING
=====

--- Checking for Null Values ---
v1      0
v2      0
dtype: int64

--- First Five Rows ---
      v1                                     v2
0  ham  Go until jurong point, crazy.. Available only ...
1  ham                Ok lar... Joking wif u oni...
2  spam  Free entry in 2 a wkly comp to win FA Cup fina...
3  ham  U dun say so early hor... U c already then say...
4  ham  Nah I don't think he goes to usf, he lives aro...

--- Last Five Rows ---
      v1                                     v2
5567 spam  This is the 2nd time we have tried 2 contact u...
5568 ham                Will Å¼ b going to esplanade fr home?
5569 ham  Pity, * was in mood for that. So...any other s...
5570 ham  The guy did some bitching but I acted like i'd...
5571 ham                Rofl. Its true to its name

--- After Renaming Columns ---
      label                                     message
0  ham  Go until jurong point, crazy.. Available only ...
1  ham                Ok lar... Joking wif u oni...
2  spam  Free entry in 2 a wkly comp to win FA Cup fina...
3  ham  U dun say so early hor... U c already then say...
4  ham  Nah I don't think he goes to usf, he lives aro...

--- Shape of Dataset ---
(5572, 2)

--- Class Distribution ---
label
ham      4825
spam      747
dtype: int64

--- Plotting Label Distribution ---
Saved: label_distribution.png

--- Preprocessing Text (NLP) ---
First 5 preprocessed texts:
1. ok lar joke wif u oni
2. free entri wkli comp win fa cup final tkt st may text fa receiv entri question std txt rate c appli
3. u dun say earli hor u c already say
4. nah think goe usf live around though
5. freemsg hey darl week word back like fun still tb ok xxx std chg send rcv

--- Creating Bag of Words Model ---
```

Email Spam Detection - Output Report

Feature Matrix Shape: (5572, 4000)

Target Vector Shape: (5572,)

--- Splitting Dataset (80% Train, 20% Test) ---

Training Set: 4457 samples

Testing Set: 1115 samples

--- Training Models ---

Training Random Forest Classifier...

Training Decision Tree Classifier...

Training Multinomial Naive Bayes...

All models trained successfully!

MODEL EVALUATION RESULTS

--- Random Forest Classifier ---

Confusion Matrix:

```
[[966   0]
 [ 25 124]]
```

Accuracy: 0.9776

--- Decision Tree Classifier ---

Confusion Matrix:

```
[[953  13]
 [ 18 131]]
```

Accuracy: 0.9722

--- Multinomial Naive Bayes ---

Confusion Matrix:

```
[[953  13]
 [   7 142]]
```

Accuracy: 0.9821

--- Confusion Matrix Heatmap (Multinomial Naive Bayes) ---

Saved: confusion_matrix_mnb.png

CLASSIFICATION REPORTS

Classification Report for Random Forest Classifier:

	precision	recall	f1-score	support
False	0.97	1.00	0.99	966
True	1.00	0.83	0.91	149
accuracy			0.98	1115
macro avg	0.99	0.92	0.95	1115
weighted avg	0.98	0.98	0.98	1115

Classification Report for Decision Tree Classifier:

	precision	recall	f1-score	support
False	0.98	0.99	0.98	966
True	0.91	0.88	0.89	149
accuracy			0.97	1115
macro avg	0.95	0.93	0.94	1115
weighted avg	0.97	0.97	0.97	1115

Email Spam Detection - Output Report

Classification Report for Multinomial Naive Bayes:

	precision	recall	f1-score	support
False	0.99	0.99	0.99	966
True	0.92	0.95	0.93	149
accuracy			0.98	1115
macro avg	0.95	0.97	0.96	1115
weighted avg	0.98	0.98	0.98	1115

=====
BEST MODEL: Multinomial Naive Bayes
=====

--- Saving Models ---

Saved: RFC.pkl (Random Forest Classifier)

Saved: DTC.pkl (Decision Tree Classifier)

Saved: MNB.pkl (Multinomial Naive Bayes)

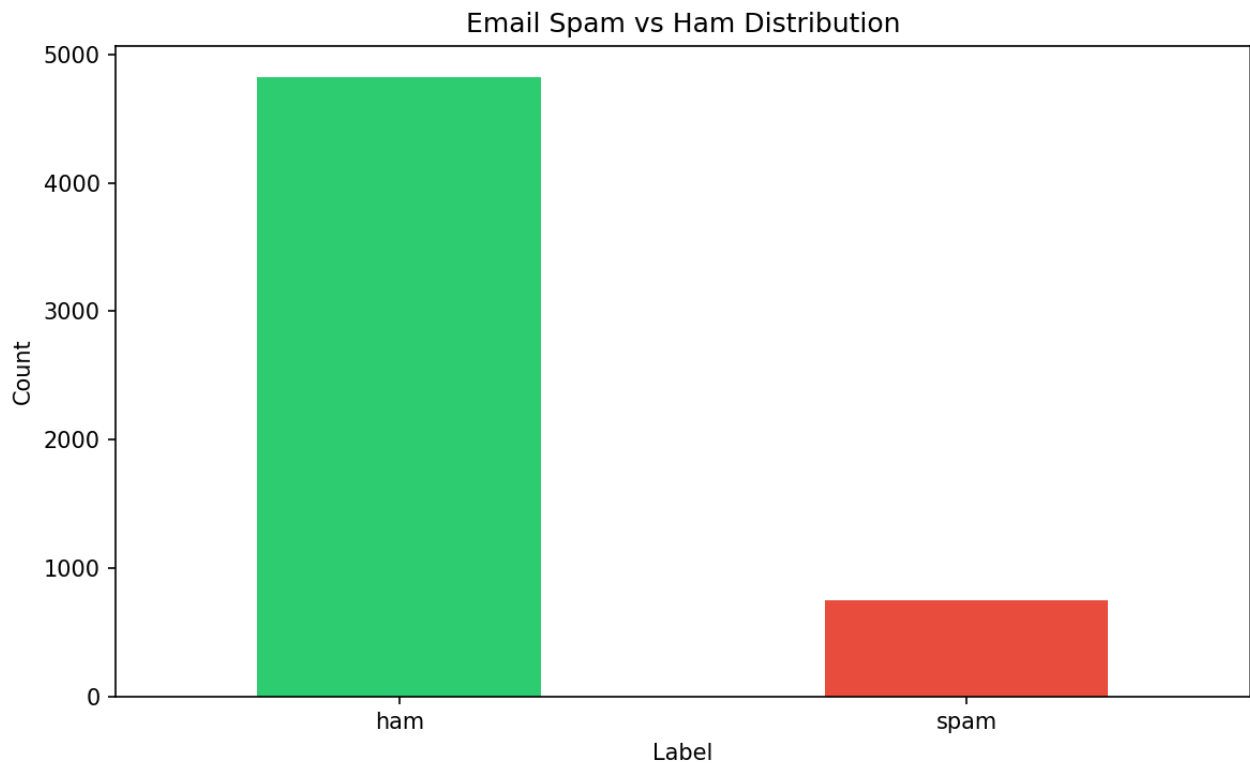
Saved: CountVectorizer.pkl

=====
ALL MODELS SAVED SUCCESSFULLY!
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Email Spam Detection - Output Report

Visualizations

Email Spam vs Ham Distribution



Email Spam Detection - Output Report

Confusion Matrix - Multinomial Naive Bayes

